

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP30020
Subject Title	From Semiconductor to Intelligent Devices
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Exclusion: AP40006
Objectives	The aim of the subject is to provide the students with an understanding of semiconductor physics, semiconductor materials properties and operation of various types of semiconductor junction diodes, transistors, integrated circuit, intelligent sensors, microactuators and memory devices required for intelligent systems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) explain the physical properties and electrical transport in drift and diffusion of various types of semiconductors; (b) explain nonequilibrium with excess carriers, and solve problems of the quasi-Fermi level splitting; (c) explain the behaviors of pn junctions under bias and the functions of various electronic and optoelectronic junction devices; (d) explain charge transport in semiconductor transistors and <i>I-V</i> characteristics. (e) apply the knowledge of semiconductors and other smart materials to explain the functions of various types of intelligent devices (f) describe the working principles of intelligent sensors, microactuators and memory devices.
Subject Synopsis/ Indicative Syllabus	Semiconductor materials and growth: structure and physical properties of semiconductor materials; crystal and epitaxial growth, energy bands Charge carriers in semiconductors: electrons and holes; intrinsic and extrinsic semiconductors; Fermi level; equation of charge carrier transport; carrier concentrations; conductivity; excess carriers; quasi-Fermi levels p-n junctions and diodes: contact potential and depletion layer; biased junction and diode equation; junction capacitance; metal-semiconductor junctions; electronic and optoelectronic devices Transistors: charge transport in a bipolar junction transistor (BJT) and metal oxide semiconductor field-effect transistor (MOSFET); <i>I-V</i> output and transfer characteristics; amplification; switching using transistors Overview of intelligent devices: integrated circuit (IC) technology; chips and computer operation; input-output interface; review of smart materials Working principles of intelligent devices: device capabilities; intelligent sensors; memory devices; microactuators; integrated intelligent device systems
Teaching/Learning Methodology	Lecture: The fundamentals of semiconductor physics, materials and various intelligent devices will be described. Students are free to request help. Students are encouraged to solve problems and to use their own knowledge to verify their solutions before seeking assistance.

	Tutorial: A set of problems and group discussion topics will be arranged in the tutorial classes. Students are encouraged to solve problems before having solutions. Laboratory: A set of laboratories /demonstrations will be provided. Students will have the opportunity to apply the fundamental knowledge gained from the lecture into practical materials test and device applications and hence develop a deeper understanding of the subject.							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	f
	1. Continuous assessment	40	✓	✓	✓	✓	✓	✓
	2. Examination	60	✓	✓	✓	✓	✓	✓
	Total	100						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Continuous assessment is based on assignments, laboratory reports, presentations, quizzes and test. The test is a one-hour written middle-term test. The examination is a 2-hour written final examination, which covers all of the key issues related to the learning outcomes.							
Student Study Effort Expected	Class contact:							
	▪ Lecture & Tutorial						32 Hrs.	
	▪ Laboratory						9 Hrs.	
	▪ Other student study effort:							
	Self-study						79 Hrs.	
	▪ Total student study effort						120 Hrs.	
Reading List and References	B.G. Streetman and S. K. Banerjee, —Solid State Electronic Devices, Prentice-Hall, 6 th Edition, 2006.							
	S.M. Sze and Kwok K. Ng, —Physics of Semiconductor Devices, John Wiley & Sons, Inc., 3 rd Edition, 2007.							
	Alan S. Morris and Reza Langari, —Measurement and Instrumentation Theory and Application, Chap. 10-Intelligent Devices, Elsevier Inc., 2 nd Edition, 2016.							
	Simon Deleonibus Ed., —Intelligent Integrated Systems Devices, Technologies, and Architectures, Jenny Stanford, 1 st Edition, 2013.							